

EXPLAINED EXAMPLE 9 – COMBINED CYCLE

- ① A small engine runs with a thermal efficiency of 0.3. Because the efficiency is so low, the ~~rejected heat~~ ^{input} is used to run a refrigeration cycle with a coefficient of performance of 4. Draw the heat engine diagram for this configuration, labeling the heat and work flows for the two devices. Then calculate the ratio of the heat input to the engine to the heat input to the refrigerator.

Given : $\eta_{th} = 0.3$

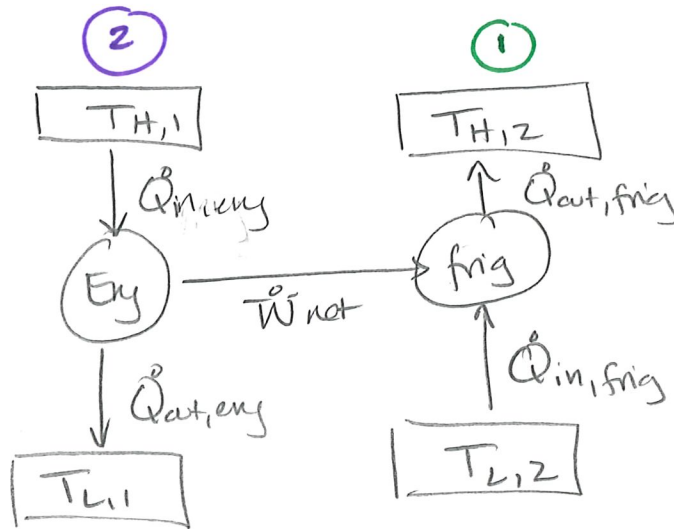
② $\beta = 4$

Find : $\frac{\dot{Q}_{in, engine}}{\dot{Q}_{in, frig}}$

Assume : steady operation

- ① : no energy losses to the outside

→ Diagram :



② efficiency : $\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in, eng}}$

COP : $\beta = \frac{\dot{Q}_{in, frig}}{\dot{W}_{net}}$

b/c point of a frig is to remove heat from a cold space

→ ratio : $\dot{W}_{net} = \eta_{th} \dot{Q}_{in,eng}$

①

$$\beta = \frac{\dot{Q}_{in,frig}}{\eta_{th} \dot{Q}_{in,eng}} \Rightarrow \eta_{th} \beta = \frac{\dot{Q}_{in,frig}}{\dot{Q}_{in,eng}}$$

$$\Rightarrow \frac{\dot{Q}_{in,eng}}{\dot{Q}_{in,frig}} = \frac{1}{\eta_{th} \beta} = \frac{1}{(0.3)(4)} \Rightarrow \boxed{0.83}$$